

Basic Framework

MC AI-REML

Writing $x = V^{-1}y$ and $K_1 = W$, $K_2 = I$, each iteration takes the average-information Newton step

$$\theta \leftarrow \theta + (\mathcal{A} + \lambda I)^{-1} s$$

with

$$s_i = \frac{1}{2}(x^T K_i x - \text{tr}(V^{-1} K_i)), \quad \mathcal{A}_{ij} = \frac{1}{2}(K_i x)^T V^{-1} (K_j x).$$

Data quadratics are exact given x : $x^T W x = x^T (W x)$, $x^T I x = \|x\|^2$; the traces use Nmc Rademacher probes $U = [u_1, \dots, u_{\text{Nmc}}] \in \mathbb{R}^{n \times \text{Nmc}}$, $\text{tr}(V^{-1} K_i) \approx \text{Nmc}^{-1} \sum_{b=1}^{\text{Nmc}} (V^{-1} u_b)^T K_i u_b$. Every V^{-1} is applied by batched CG whose only primitive is the V mat-vec

$$VU = \sigma_{\text{gsg}}^2(WU) + \sigma_e^2 U.$$

WU is the standardized linear operator used in MoM.

1. Original method (extended from single-gene case): $O(nm^2)$

This method build linear operator Wu implicitly using one random vector (matrix) per apply.

Setup (once per gene). From Z :

1. the weight matrices $B, R, T \in \mathbb{R}^{m \times m}$, symmetric with zero diagonal

$$B_{ij} = \frac{1}{\sigma_{ij}^2}, \quad R_{ij} = \frac{\mu_{ij}}{\sigma_{ij}^2}, \quad T_{ij} = \frac{\mu_{ij}^2}{\sigma_{ij}^2}$$

where empirical $\mu_{ij} = \mathbb{E}[Z_i Z_j]$ and $\sigma_{ij}^2 = \text{Var}(Z_i Z_j) - O(nm^2)$

1. the vector $v_R = (Z \circ (ZR)) \mathbf{1} - O(nm^2)$;
2. the scalar $s_T = \mathbf{1}^T T \mathbf{1} \in \mathbb{R} - O(m^2)$.

Only Z , V , v_R and s_T stay; R , T and the $n \times m$ intermediate ZR can all be discarded. **Apply (per vector u).** Exactly one u -dependent object is built:

$$s_u = \mathbf{1}^T u, \quad M = Z^T \text{diag}(u) Z$$

and the result is

$$Wu = \frac{1}{2p} [(Z \circ (Z(B \circ M))) \mathbf{1} - s_u v_R + (s_u s_T - u^T v_R) \mathbf{1}].$$

1.1 Optimization: can we do it using LD score rather than weight matrices

LD Score

Let the reference panel contain n individuals and m SNPs. For variant i , let p_i denote the allele frequency and $g_i \in \{0, 1, 2\}$ the raw genotype count. The standardized genotype is

$$Z_i = \frac{g_i - 2p_i}{\sqrt{2p_i(1 - p_i)}},$$

so that $\mathbb{E}[Z_i] = 0$ and $\text{Var}(Z_i) = 1$. LD between variants i and j is measured by the genotypic correlation

$$r_{ij} = \text{Cor}(g_i, g_j) = \mathbb{E}[Z_i Z_j],$$

and we write $R \in \mathbb{R}^{m \times m}$ for the LD correlation matrix.

The LD score of SNP i is the sum of squared correlations with all variants:

$$\ell_i = \sum_{j=1}^m r_{ij}^2 = \sum_{j=1}^m \mathbb{E}[Z_i Z_j]^2 = \sum_{j=1}^m (R \circ R)_{ij} = \sum_{j=1}^m \mu_{ij}^2$$

- link LD score to $\mu_{ij} = \mathbb{E}[Z_i Z_j]$ and $\sigma_{ij}^2 = \text{Var}(Z_i Z_j)$:

$$\ell_i = \sum_{j=1}^m \mu_{ij}^2$$

$$\sigma_{ij}^2 = \mathbb{E}[Z_i^2 Z_j^2] - \mu_{ij}^2 \iff \mathbb{E}[Z_i^2 Z_j^2] = \sigma_{ij}^2 + \mu_{ij}^2$$

$$\ell_i = \sum_{j=1}^m (\mathbb{E}[Z_i^2 Z_j^2] - \sigma_{ij}^2)$$

$$\sum_{ij} \mu_{ij}^2 = \sum_i \ell_i$$

$$\sum_{i \neq j} \mu_{ij}^2 = \sum_{i=1}^m \ell_i - m$$

$$\sum_{i \neq j} \sigma_{ij}^2 = ?$$

2 Different way to build linear operator (remove $\circ$), can we do it in $O(nm)$?

make it easy, start from unstandardized way:

$$\begin{aligned} W &= p^{-1} H H^T \\ &= \frac{1}{2p} ((Z Z^T) \circ (Z Z^T) - (Z \circ Z)(Z \circ Z)^T) \\ &= \frac{1}{2p} (K_w \circ K_w - D D^T) \end{aligned}$$

$(Z \circ Z)(Z \circ Z)^T$ is easy, $Z \circ Z$ takes $O(mn)$ and same space like Z , The linear operator is $(Z \circ Z)(Z \circ Z)^T u$, takes $O(mn)$.

Then we should use way to remove $\circ$ in $(Z Z^T) \circ (Z Z^T)$. Some useful identities:

$$u^T (A \circ B) v = \text{tr}(D_u A D_v B^T)$$

$$(A \circ B) v = \text{diag}(A D_v B^T)$$

$$(A \circ B) u = E[(A D_u B^T z) \circ z]$$

The linear operator:

$$\begin{aligned}
((\mathbf{Z}\mathbf{Z}^T) \circ (\mathbf{Z}\mathbf{Z}^T))u &= \text{diag}(\mathbf{Z}\mathbf{Z}^T D_u \mathbf{Z}\mathbf{Z}^T) \\
&= \mathbb{E}[\left((\mathbf{Z}\mathbf{Z}^T) D_u (\mathbf{Z}\mathbf{Z}^T) v\right) \circ v] \\
&\approx \frac{1}{N_{\text{mc}}} \sum_{i=1}^{N_{\text{mc}}} (\mathbf{Z}\mathbf{Z}^T D_u \mathbf{Z}\mathbf{Z}^T v_i) \circ v_i \\
&= \frac{1}{N_{\text{mc}}} ((\mathbf{Z}(\mathbf{Z}^T(u \circ (\mathbf{Z}(\mathbf{Z}^T \mathbf{V})))) \circ \mathbf{V}) \mathbf{1}_{N_{\text{mc}}})
\end{aligned}$$

where v is a random vector and V is the matrix form. This is $O(mn N_{\text{mc}})$. Overall,

$$\begin{aligned}
Wu &= \frac{1}{2p} [((\mathbf{K}_w \circ \mathbf{K}_w)u - \mathbf{D}(\mathbf{D}^T u)] \\
&\approx \frac{1}{2p} \left[\frac{1}{N_{\text{mc}}} ((\mathbf{Z}(\mathbf{Z}^T(u \circ (\mathbf{Z}(\mathbf{Z}^T \mathbf{V})))) \circ \mathbf{V}) \mathbf{1}_{N_{\text{mc}}} - \mathbf{D}(\mathbf{D}^T u) \right]
\end{aligned}$$

where $\mathbf{D} = \mathbf{Z} \circ \mathbf{Z}$ and $\mathbf{V} = [v_1, \dots, v_{N_{\text{mc}}}] \in \mathbb{R}^{n \times N_{\text{mc}}}$ has i.i.d. Rademacher entries. The whole apply costs $O(mn N_{\text{mc}})$ against $O(nm^2)$ for the exact route.

Then, we should do standardized Wu , refer the $O(nm^2)$ exact linear operator:

$$Wu = \frac{1}{2p} [(Z \circ (Z(B \circ M)))\mathbf{1} - s_u v_R + (s_u s_T - u^T v_R)\mathbf{1}].$$

can be separated as:

$$Wu = \underbrace{\frac{1}{2p} [(Z \circ (ZM))\mathbf{1} - \mathbf{D}(\mathbf{D}^T u)]}_{\text{raw}} + \underbrace{\frac{1}{2p} [-s_u v_R + (s_u s_T - u^T v_R)\mathbf{1}]}_{\text{std staff}}$$

1. the vector $v_R = (Z \circ (ZR))\mathbf{1} - O(nm^2)$;
2. the scalar $s_T = \mathbf{1}^T T \mathbf{1} \in \mathbb{R} - O(m^2)$.

$$\hat{v}_R = (Z \circ (ZR))\mathbf{1} = \frac{1}{n} (Z \circ (Z(\mathbf{Z}^T \mathbf{Z})))\mathbf{1} - (Z \circ Z)\mathbf{1} \approx \frac{1}{n N_{\text{mc}}} ((\mathbf{Z}(\mathbf{Z}^T(\mathbf{Z}(\mathbf{Z}^T \mathbf{V})))) \circ \mathbf{V}) \mathbf{1}_{N_{\text{mc}}} - (Z \circ Z)\mathbf{1}$$

$$\hat{s}_T = \frac{1}{n} \mathbf{1}^T \hat{v}_R$$

Finally, the std linear operator for Wu is

$$\begin{aligned}
Wu &= \frac{1}{2p} [(\mathbf{K}_w \circ \mathbf{K}_w)u - \mathbf{D}(\mathbf{D}^T u)] + \frac{1}{2p} [-s_u v_R + (s_u s_T - u^T v_R)\mathbf{1}] \\
&\approx \frac{1}{2p} \left[\frac{1}{N_{\text{mc}}} ((\mathbf{Z}(\mathbf{Z}^T(u \circ (\mathbf{Z}(\mathbf{Z}^T \mathbf{V})))) \circ \mathbf{V}) \mathbf{1}_{N_{\text{mc}}} - \mathbf{D}(\mathbf{D}^T u) \right] + \frac{1}{2p} [-s_u \hat{v}_R + (s_u \hat{s}_T - u^T \hat{v}_R)\mathbf{1}]
\end{aligned}$$

where, with the same probe matrix $\mathbf{V} \in \mathbb{R}^{n \times N_{\text{mc}}}$ reused at $u = \mathbf{1}$,

$$\hat{v}_R = \frac{1}{n N_{\text{mc}}} ((\mathbf{Z}(\mathbf{Z}^T(\mathbf{Z}(\mathbf{Z}^T \mathbf{V})))) \circ \mathbf{V}) \mathbf{1}_{N_{\text{mc}}} - (Z \circ Z)\mathbf{1}, \quad \hat{s}_T = \frac{1}{n} \mathbf{1}^T \hat{v}_R.$$

We can do MoM in $O(nm)$

We estimate $\text{tr}[W]$, $\text{tr}[W^2]$ and $y^T W y$ without forming K , using the stochastic trace estimator

$$\hat{\text{tr}}[A] = \frac{1}{n_{mc}} \sum_{l=1}^{n_{mc}} u_l^T A u_l,$$

where $n_{mc} \ll \min(n, m)$ and each u_l is a random vector with independent, mean-zero, unit-variance elements. We denote this class by U .

Pair probes. For each row $t = 1, \dots, n$ and $u \in U^m$,

$$u^T X_{t.}^T X_{t.} u = 2 \sum_{i < j}^m X_{ti} X_{tj} u_i u_j + \sum_i^m X_{ti}^2 u_i^2 = 2H_{t.} v + X_{t.} X_{t.}^T,$$

where $v_k = u_i u_j$ for $k \leftrightarrow (i < j)$ satisfies $v \in U^p$. Hence $z = Hv$ is available elementwise from a single Xu :

$$z_t = \frac{1}{2}(u^T X_{t.}^T X_{t.} u - X_{t.} X_{t.}^T), \quad O(nm).$$

Since $\mathbb{E}[vv^T] = I_p$, we have $\mathbb{E}[zz^T] = HH^T = pW$, so one z serves both traces.

$\text{tr}[W]$

$$\text{tr}[W] = p^{-1} \mathbb{E}[z^T z] = \frac{1}{4p} \mathbb{E} \sum_t (u^T X_{t.}^T X_{t.} u - X_{t.} X_{t.}^T)^2,$$

$$\hat{\text{tr}}[W] = \frac{1}{4pn_{mc}} \sum_{l=1}^{n_{mc}} \sum_{t=1}^n (\tilde{u}_l^T X_{t.}^T X_{t.} \tilde{u}_l - X_{t.} X_{t.}^T)^2, \quad \tilde{u}_l \in U^m.$$

$\text{tr}[W^2]$

$$\begin{aligned} \text{tr}[W^2] &= p^{-2} \mathbb{E}[z^T H H^T z] = p^{-2} \mathbb{E} \sum_{i < j} \left(\sum_{t=1}^n X_{ti} X_{tj} z_t \right)^2 \\ &= \frac{1}{2p^2} \left(\sum_{i,j} \sum_{t,l} X_{li} X_{lj} X_{ti} X_{tj} z_t z_l - \sum_{i=1}^m \sum_{t,l} X_{li}^2 X_{ti}^2 z_t z_l \right) \\ &= \frac{1}{2p^2} \text{tr}[K D_z K D_z] - \frac{1}{2p^2} z^T (X \circ X)(X \circ X)^T z, \end{aligned}$$

where D_z is the diagonal matrix generated by z . Averaging pair probes $u_l \in U^m$,

$$\hat{z}_t = \frac{1}{2\sqrt{n_{mc}}} \sum_{l=1}^{n_{mc}} (u_l^T X_{t.}^T X_{t.} u_l - X_{t.} X_{t.}^T),$$

the $\sqrt{n_{mc}}$ keeping $\mathbb{E}[\hat{z}\hat{z}^T] = pW$. Given $\hat{z}$,

$$\hat{\text{tr}}[W^2] = \frac{1}{2p^2 n_{mc}} \sum_{l=1}^{n_{mc}} \tilde{u}_l^T X X^T D_{\hat{z}} X X^T D_{\hat{z}} \tilde{u}_l - \frac{1}{2p^2} \hat{z}^T (X \circ X)(X \circ X)^T \hat{z}.$$

Every term above is applied right to left, so $K = XX^T$ is never formed and each probe costs four matrix–vector products with X , i.e. $O(nm)$.

how to solve that

The cross terms cannot be dropped. Each W_g is positive semidefinite, so $\text{tr}[W_g W_h] = \angle(W_g, W_h)_F \geq 0$, and since $\mathbb{E}[W_g] = I_n$ for standardized genotypes, independence between two

genes gives $\mathbb{E} \operatorname{tr}[W_g W_h] = \operatorname{tr}[I_n] = n$ rather than zero. Their total therefore grows like $G(G-1)n$ against Gn for the diagonal part, so they dominate $\operatorname{tr}[W^2]$ for any realistic G .

They can, however, be made implicit. Draw one pair probe $u_g \in U^{m_g}$ per gene, **independently across genes**, and form

$$z_{g,t} = \frac{1}{2\sqrt{p_g}}(u_g^T X_{g,t}^T X_{g,t} u_g - X_{g,t} X_{g,t}^T), \quad z = \sum_{g=1}^G z_g.$$

Then $\mathbb{E}[z_g z_g^T] = p_g^{-1} H_g H_g^T = W_g$, and independence kills every $\mathbb{E}[z_g z_h^T]$, so

$$\mathbb{E}[z z^T] = \sum_{g=1}^G W_g = W \implies \operatorname{tr}[W^2] = \mathbb{E}[z^T W z] = \mathbb{E}\left[\sum_{g=1}^G z^T W_g z\right].$$

The cross terms are carried by the covariance of z and never have to be enumerated.

The same z must enter every term. Expanding $z^T W_g z = \sum_{h,h'} z_h^T W_g z_{h'}$ shows where they live; replacing z by the gene's own z_g in the g -th term would return $\sum_g \operatorname{tr}[W_g^2]$ and discard exactly the part that dominates.

Each summand is then the single-gene quadratic form, evaluated with an inner probe $c_l \in U^{m_g}$ drawn separately per gene:

$$z^T W_g z = \frac{1}{2p_g} \left(\operatorname{tr}[M_{g,z}^2] - \|D_g^T z\|^2 \right), \quad M_{g,z} = X_g^T D_z X_g,$$

$$\operatorname{tr}[M_{g,z}^2] \approx \frac{1}{n_{\text{in}}} \sum_{l=1}^{n_{\text{in}}} \|X_g^T (z \circ (X_g c_l))\|^2.$$

Since $\mathbb{E}[z z^T] = W$ already holds for a single draw, enlarging the probe count inside one z does not reduce the variance; the estimator must be averaged over R independent draws,

$$\hat{\operatorname{tr}}[W^2] = \frac{1}{R} \sum_{r=1}^R \sum_{g=1}^G \left(\frac{1}{2p_g n_{\text{in}}} \sum_{l=1}^{n_{\text{in}}} \|X_g^T (z^{(r)} \circ (X_g c_l))\|^2 - \frac{1}{2p_g} \|D_g^T z^{(r)}\|^2 \right),$$

converging at $O(R^{-\frac{1}{2}})$. Building one z costs $O(nM)$ and contracting it costs $O(nM n_{\text{in}})$, so the total is $O(nMR n_{\text{in}})$ – linear in the number of variants and with no $\binom{G}{2}$ term anywhere.

how to solve that

Draw an independent pair probe $u_g \in U^{m_g}$ per gene and set

$$z_{g,t} = \frac{1}{2\sqrt{p_g}}(u_g^T X_{g,t}^T X_{g,t} u_g - X_{g,t} X_{g,t}^T), \quad z = \sum_{g=1}^G z_g,$$

so that $\mathbb{E}[z_g z_g^T] = p_g^{-1} H_g H_g^T = W_g$ and, by independence across genes,

$$\mathbb{E}[z z^T] = \sum_{g=1}^G W_g = W \implies \operatorname{tr}[W^2] = \mathbb{E}[z^T W z] = \mathbb{E}\left[\sum_{g=1}^G z^T W_g z\right].$$

The cross terms never appear: they are carried by the covariance of z , and the quadratic form is evaluated gene by gene with the single-gene machinery,

$$z^T W_g z = \frac{1}{2p_g} \left(\text{tr}[M_{g,z}^2] - \|D_g^T z\|^2 \right), \quad M_{g,z} = X_g^T D_z X_g, \quad \text{tr}[M_{g,z}^2] \approx \frac{1}{n_{\text{in}}} \sum_{l=1}^{n_{\text{in}}} \|X_g^T (z \circ (X_g c_l))\|^2,$$

with $c_l \in U^{m_g}$. Independence of the u_g is essential: reusing one probe across genes would make $\mathbb{E}[z_g z_h^T] \neq 0$ and bias the estimate.

Averaging

$\mathbb{E}[zz^T] = W$ holds for a single draw, so raising the number of pair probes inside one z does not reduce the variance of $\text{tr}[W^2]$; the estimator must be averaged over R independent draws of z :

$$\hat{\text{tr}}[W^2] = \frac{1}{R} \sum_{r=1}^R \sum_{g=1}^G \left(\frac{1}{2p_g n_{\text{in}}} \sum_{l=1}^{n_{\text{in}}} \|X_g^T (z^{(r)} \circ (X_g c_l))\|^2 - \frac{1}{2p_g} \|D_g^T z^{(r)}\|^2 \right),$$

which converges at $O(R^{-\frac{1}{2}})$. Each draw costs $O(nM)$ to build and $O(nMn_{\text{in}})$ to contract, so the total is $O(nMRn_{\text{in}})$ with no dependence on G beyond M .

Per-gene variance components

If instead each gene carries its own σ_g^2 , the moment equations need the full $G \times G$ matrix $\text{tr}[W_g W_h]$, which the shared probe cannot deliver — it returns only the sum. Drawing G separate probes and using $\text{tr}[W_g W_h] = \mathbb{E}[z_g^T W_h z_g]$ restores them at $\binom{G}{2}$ quadratic forms, which is the practical reason a single pooled component is usually estimated instead.

2.3 SVD

The rank-one identity. For vectors $a, c \in \mathbb{R}^n$ and $b, d \in \mathbb{R}^m$,

$$(ab^T) \circ (cd^T) = (a \circ c)(b \circ d)^T,$$

since entry (i, j) reads $a_i b_j c_i d_j$ on the left and $(a_i c_i)(b_j d_j)$ on the right. **A Hadamard product of rank-one matrices is rank one**, and its factors are the elementwise products of the original factors.

General form. Take low-rank factorizations $A \approx L_A R_A$ and $B \approx L_B R_B$, where $L_A \in \mathbb{R}^{n \times r}$, $R_A \in \mathbb{R}^{r \times n}$ retain the r leading singular values of A and likewise k for B . Write $l_{As} \in \mathbb{R}^n$ for column s of L_A and $\rho_{As} \in \mathbb{R}^n$ for row s of R_A transposed, and define

$$w_{st} := l_{As} \circ l_{Bt} \in \mathbb{R}^n, \quad z_{st} := \rho_{As} \circ \rho_{Bt} \in \mathbb{R}^n.$$

Expanding both sums and applying the identity term by term,

$$A \circ B \approx \sum_{s=1}^r \sum_{t=1}^k w_{st} z_{st}^T, \quad \text{so} \quad (A \circ B)u \approx \sum_{s=1}^r \sum_{t=1}^k w_{st} (z_{st}^T u).$$

Each term is one inner product times one vector, so no $n \times n$ object is ever formed and the apply costs $O(nrk)$. Nothing here needs A or B beyond their factors.

Specialization to $K_w \circ K_w$. Two features of the present case collapse the double sum. First, $K_w = ZZ^T$ is symmetric positive semidefinite, so its SVD coincides with its eigendecomposition and the left and right factors agree. Taking the thin SVD $Z = U\Sigma V^T$ with $Z \in \mathbb{R}^{n \times m}$,

$$K_w = U\Sigma^2 U^T = \sum_{s=1}^m \lambda_s q_s q_s^T, \quad \lambda_s = \sigma_s^2, \quad q_s = u_s \in \mathbb{R}^n,$$

so the eigenpairs come from the SVD of the $n \times m$ matrix $\mathbf{Z}$ – in practice from the $m \times m$ Gram $\mathbf{Z}^T \mathbf{Z} = \mathbf{V} \mathbf{\Sigma}^2 \mathbf{V}^T$ followed by $\mathbf{U} = \mathbf{Z} \mathbf{V} \mathbf{\Sigma}^{-1}$, never from $\mathbf{K}_w$ itself. Second, only **one** of the two factors needs expanding. Writing $[\mathbf{K}_w \circ \mathbf{K}_w]_{ij} = K_{ij} \cdot K_{ij}$ and substituting the layer sum into the first copy alone,

$$[(\mathbf{K}_w \circ \mathbf{K}_w)u]_i = \sum_j \left(\sum_s \lambda_s q_{si} q_{sj} \right) K_{ij} u_j = \sum_s \lambda_s q_{si} [\mathbf{K}_w(q_s \circ u)]_i,$$

that is

$$(\mathbf{K}_w \circ \mathbf{K}_w)u = \sum_{s=1}^r \lambda_s q_s \circ (\mathbf{K}_w(q_s \circ u)) = \sum_{s=1}^r \lambda_s q_s \circ (\mathbf{Z}(\mathbf{Z}^T(q_s \circ u))).$$

The retained copy of $\mathbf{K}_w$ is applied as two products with $\mathbf{Z}$, so each of the r terms costs $O(nm)$ and the apply is $O(nmr)$ rather than $O(nr^2)$. The full operator is

$$\mathbf{W}u = \frac{1}{2p} \left[\sum_{s=1}^r \lambda_s q_s \circ (\mathbf{Z}(\mathbf{Z}^T(q_s \circ u))) - \mathbf{D}(\mathbf{D}^T u) \right], \quad \mathbf{D} = \mathbf{Z} \circ \mathbf{Z}.$$